

Assignment 9: Time Series Analysis

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A09_TimeSeries.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Check your working directory
 - Load the tidyverse, zoo, and trend packages
 - Set your ggplot theme

```
#loading packages
library(tidyverse)
library(zoo)
library(trend)
library(here)
library(Kendall)

#working directory
getwd()
here()

#building theme
mytheme <- theme_minimal(base_size = 12) +
  theme(axis.text = element_text(color = "darkgrey"), legend.position = "right",
        title = element_text(color = "black"))
theme_set(mytheme)
```

2. Import the ten datasets from the Ozone_TimeSeries folder in the Raw data folder. These contain ozone concentrations at Garinger High School in North Carolina from 2010-2019 (the EPA air database only allows downloads for one year at a time). Import these either individually or in bulk and then combine them into a single dataframe named `GaringerOzone` of 3589 observation and 20 variables.

```
#1
#reading in data
EPAair_2010 <- read.csv(
  here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2010_raw.csv"),
  stringsAsFactors = TRUE)

EPAair_2011 <- read.csv(
  here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2011_raw.csv"),
  stringsAsFactors = TRUE)

EPAair_2012 <- read.csv(
  here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2012_raw.csv"),
  stringsAsFactors = TRUE)

EPAair_2013 <- read.csv(
  here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2013_raw.csv"),
  stringsAsFactors = TRUE)

EPAair_2014 <- read.csv(
  here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2014_raw.csv"),
  stringsAsFactors = TRUE)

EPAair_2015 <- read.csv(
  here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2015_raw.csv"),
  stringsAsFactors = TRUE)

EPAair_2016 <- read.csv(
  here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2016_raw.csv"),
  stringsAsFactors = TRUE)

EPAair_2017 <- read.csv(
  here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2017_raw.csv"),
  stringsAsFactors = TRUE)

EPAair_2018 <- read.csv(
  here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2018_raw.csv"),
  stringsAsFactors = TRUE)

EPAair_2019 <- read.csv(
  here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2019_raw.csv"),
  stringsAsFactors = TRUE)

#wrangling into one processed data set
GaringerOzone <- rbind(EPAair_2010, EPAair_2011, EPAair_2012,
  EPAair_2013, EPAair_2014, EPAair_2015,
  EPAair_2016, EPAair_2017, EPAair_2018, EPAair_2019)
```

Wrangle

3. Set your date column as a date class.
4. Wrangle your dataset so that it only contains the columns Date, Daily.Max.8.hour.Ozone.Concentration, and DAILY_AQI_VALUE.
5. Notice there are a few days in each year that are missing ozone concentrations. We want to generate a daily dataset, so we will need to fill in any missing days with NA. Create a new data frame that contains a sequence of dates from 2010-01-01 to 2019-12-31 (hint: `as.data.frame(seq())`). Call this new data frame Days. Rename the column name in Days to “Date”.
6. Use a `left_join` to combine the data frames. Specify the correct order of data frames within this function so that the final dimensions are 3652 rows and 3 columns. Call your combined data frame GaringerOzone.

```
#3 #setting date column as date class
GaringerOzone$Date <- mdy(GaringerOzone$Date)

#4 #wrangling
GaringerOzone_Select <- select(GaringerOzone,
                              'Date', 'Daily.Max.8.hour.Ozone.Concentration',
                              'DAILY_AQI_VALUE')

#5
#creating data frame
Days <- as.data.frame(seq(from = as.Date("2010-01-01"),
                          to = as.Date("2019-12-31"), by = "day"))

#renaming column
Days <- rename(Days, 'Date' = 'seq(from = as.Date("2010-01-01"), to = as.Date("2019-12-31"), by = "day")')

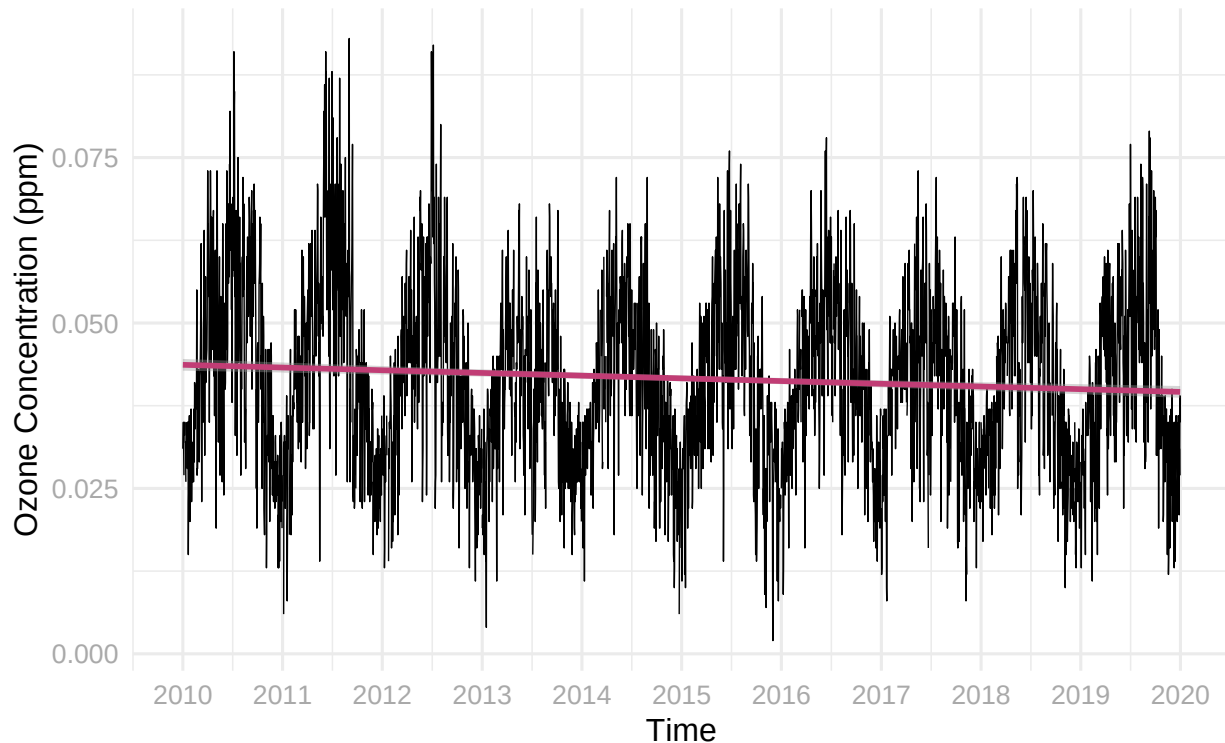
#6 #joining
GaringerOzone <- left_join(Days, GaringerOzone_Select)
```

Visualize

7. Create a line plot depicting ozone concentrations over time. In this case, we will plot actual concentrations in ppm, not AQI values. Format your axes accordingly. Add a smoothed line showing any linear trend of your data. Does your plot suggest a trend in ozone concentration over time?

```
#7
plot1 <- GaringerOzone %>%
  ggplot(aes(x = Date,
             y = Daily.Max.8.hour.Ozone.Concentration)) +
  geom_line(size = 0.25) +
  geom_smooth(method = "lm", color = "#c13d75ff") +
  scale_x_date(date_breaks = "1 year",
              date_labels = "%Y") +
  labs(x = "Time",
       y = "Ozone Concentration (ppm)",
       title = "Plotting Daily Ozone Concentration Over Time
               at Garinger High School, North Carolina")
print(plot1)
```

Plotting Daily Ozone Concentration Over Time at Garinger High School, North Carolina



Answer: This plot depicts a seasonal cycle that repeats at regular intervals over time. The trend line appears to shift downward over time, however, additional analysis is needed to determine if this downward slope is statistically significant.

Time Series Analysis

Study question: Have ozone concentrations changed over the 2010s at this station?

8. Use a linear interpolation to fill in missing daily data for ozone concentration. Why didn't we use a piecewise constant or spline interpolation?

```
#8
#linear interpolation
GaringerOzone$Daily.Max.8.hour.Ozone.Concentration <-
  na.approx(GaringerOzone$Daily.Max.8.hour.Ozone.Concentration)
```

Answer: The linear interpolation is suitable for this data because the data clearly follows cyclical pattern that would warrant a "connect the dots" approach. Spline interpolation is more suitable for preserving a curve, and piecewise interpolation is better suited for larger and more complex datasets.

9. Create a new data frame called `GaringerOzone.monthly` that contains aggregated data: mean ozone concentrations for each month. In your pipe, you will need to first add columns for year and month to form the groupings. In a separate line of code, create a new Date column with each month-year combination being set as the first day of the month (this is for graphing purposes only)

```

#9
#adding month and year columns
GaringerOzone.monthly <- GaringerOzone %>%
  mutate(Month = month(Date),
         Year = year(Date)) %>%
#group by month and year
  group_by(Year, Month) %>%
#aggregate monthly mean
  summarize(Monthly_Mean_Concentration =
            mean(Daily.Max.8.hour.Ozone.Concentration)) %>%
#adjust date column
  mutate(Date = make_date(year = Year, month = Month, day = 1)) %>%
#select relevant columns
  select('Date', 'Monthly_Mean_Concentration')

```

10. Generate two time series objects. Name the first `GaringerOzone.daily.ts` and base it on the dataframe of daily observations. Name the second `GaringerOzone.monthly.ts` and base it on the monthly average ozone values. Be sure that each specifies the correct start and end dates and the frequency of the time series.

```

#10
#daily time series
GaringerOzone.daily.ts <- ts(GaringerOzone$Daily.Max.8.hour.Ozone.Concentration,
                             start = c(2010, 1), frequency = 365)

#monthly time series
GaringerOzone.monthly.ts <- ts(GaringerOzone.monthly$Monthly_Mean_Concentration,
                               start = c(2010, 1), frequency = 12)

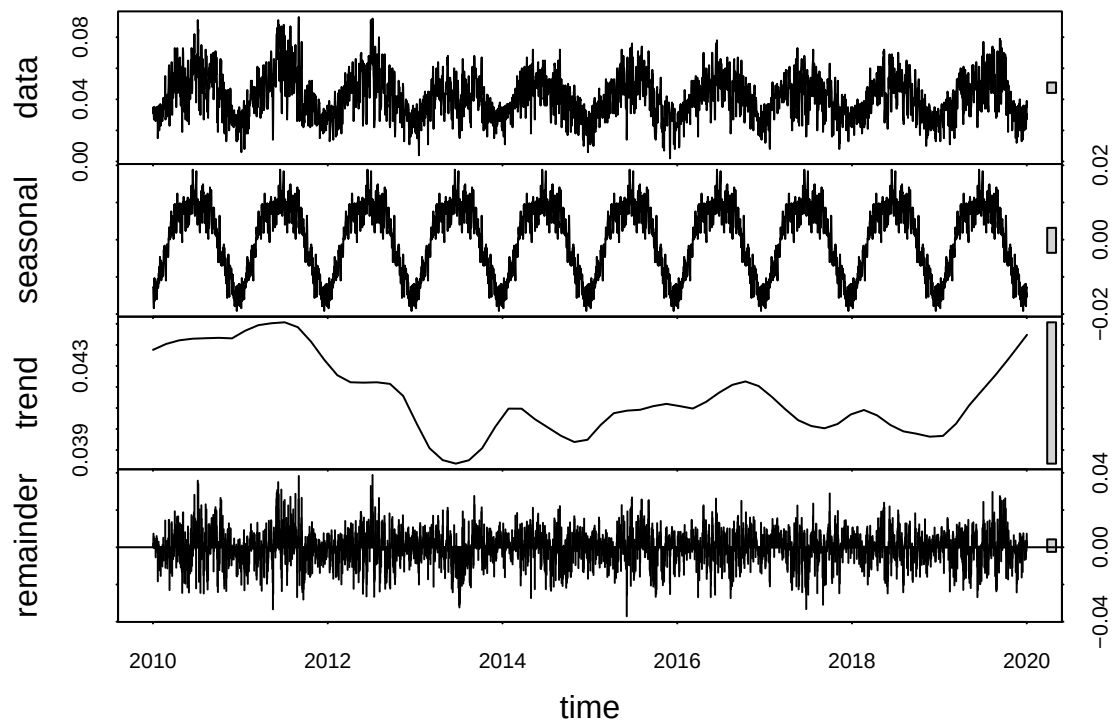
```

11. Decompose the daily and the monthly time series objects and plot the components using the `plot()` function.

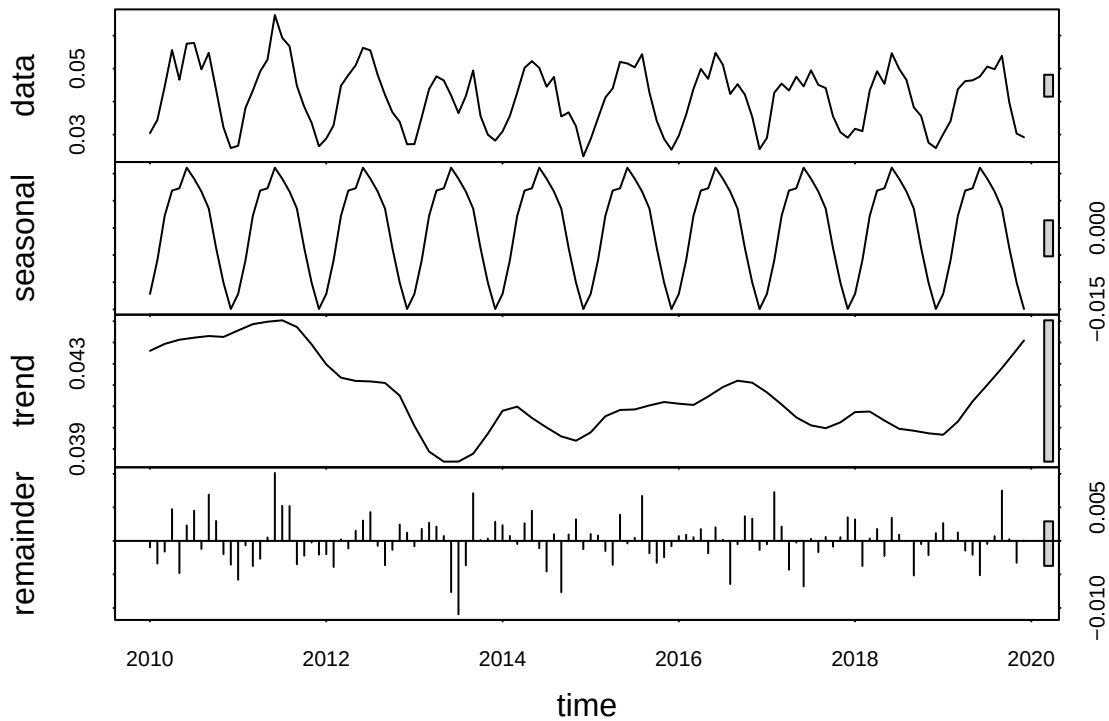
```

#11
daily_decomp <- stl(GaringerOzone.daily.ts, s.window = "periodic")
plot(daily_decomp)

```



```
monthly_decomp <- stl(GaringerOzone.monthly.ts, s.window = "periodic")
plot(monthly_decomp)
```



12. Run a monotonic trend analysis for the monthly Ozone series. In this case the seasonal Mann-Kendall is most appropriate; why is this?

```
#12
monthly_trend1 <- smk.test(GaringerOzone.monthly.ts)
summary(monthly_trend1)
```

```
##
## Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: GaringerOzone.monthly.ts
## alternative hypothesis: two.sided
##
## Statistics for individual seasons
##
## H0
##
```

	S	varS	tau	z	Pr(> z)	
## Season 1:	S = 0	15	125	0.333	1.252	0.21050
## Season 2:	S = 0	-1	125	-0.022	0.000	1.00000
## Season 3:	S = 0	-4	124	-0.090	-0.269	0.78762
## Season 4:	S = 0	-17	125	-0.378	-1.431	0.15241
## Season 5:	S = 0	-15	125	-0.333	-1.252	0.21050
## Season 6:	S = 0	-17	125	-0.378	-1.431	0.15241
## Season 7:	S = 0	-11	125	-0.244	-0.894	0.37109
## Season 8:	S = 0	-7	125	-0.156	-0.537	0.59151

```
## Season 9:   S = 0   -5  125 -0.111 -0.358  0.72051
## Season 10:  S = 0 -13  125 -0.289 -1.073  0.28313
## Season 11:  S = 0 -13  125 -0.289 -1.073  0.28313
## Season 12:  S = 0  11  125  0.244  0.894  0.37109
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
monthly_trend2 <- SeasonalMannKendall(GaringerOzone.monthly.ts)
summary(monthly_trend2)
```

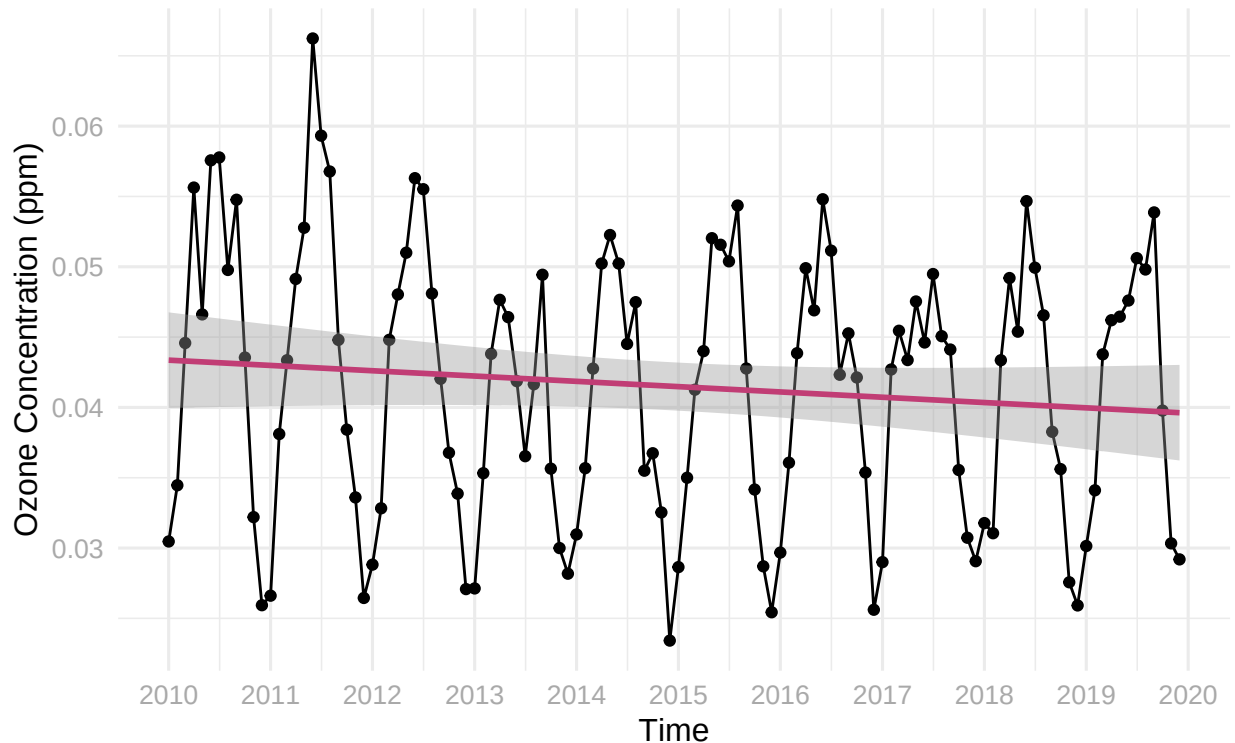
```
## Score = -77 , Var(Score) = 1499
## denominator = 539.4972
## tau = -0.143, 2-sided pvalue =0.046724
```

Answer: Seasonal Mann-Kendall is the most appropriate for this trend analysis because we know from the first plot that the data follows a seasonal pattern. None of the other tests for trend analysis incorporate seasonality, and thus are not suited to this data.

13. Create a plot depicting mean monthly ozone concentrations over time, with both a `geom_point` and a `geom_line` layer. Edit your axis labels accordingly.

```
# 13
monthly_plot <- GaringerOzone.monthly %>%
  ggplot(aes(x = Date,
             y = Monthly_Mean_Concentration)) +
  geom_point() +
  geom_line() +
  geom_smooth(method = lm, color = "#c13d75ff") +
  scale_x_date(date_breaks = "1 year",
              date_labels = "%Y") +
  labs(x = "Time",
       y = "Ozone Concentration (ppm)",
       title = "Plotting Mean Monthly Ozone Concentration Over Time
               at Garinger High School, North Carolina")
print(monthly_plot)
```


Plotting Mean Monthly Ozone Concentration Over Time at Garinger High School, North Carolina



14. To accompany your graph, summarize your results in context of the research question. Include output from the statistical test in parentheses at the end of your sentence. Feel free to use multiple sentences in your interpretation.

Answer: Based on the results of the Seasonal Mann-Kendall test, we can reject the null hypothesis that the data is ozone concentration data is randomly ordered and accept the alternative hypothesis that mean ozone concentration at this station has exhibited statistically significant change over time. The tau and score values are negative, thus confirming our previous observation that the ozone concentrations are decreasing over time.

(Score = -77 , Var(Score) = 1499

denominator = 539.4972

tau = -0.143, 2-sided pvalue =0.046724)

15. Subtract the seasonal component from the `GaringerOzone.monthly.ts`. Hint: Look at how we extracted the series components for the `EnoDischarge` on the lesson Rmd file.
16. Run the Mann Kendall test on the non-seasonal Ozone monthly series. Compare the results with the ones obtained with the Seasonal Mann Kendall on the complete series.

```

#15
GaringerOzone.monthly.components <- as.data.frame(
  monthly_decomp$time.series[,1:3])

GaringerOzone.monthly.components <- GaringerOzone.monthly.components %>%
  mutate(Observed = GaringerOzone.monthly$Monthly_Mean_Concentration,
         Date = GaringerOzone.monthly$Date)

#subtracting
GaringerOzone.monthly.components <- GaringerOzone.monthly.components %>%
  mutate(Observed_No_Season = GaringerOzone.monthly.components$trend +
         GaringerOzone.monthly.components$remainder)
#trend + remainder = observed - seasonal

#16
non_seasonal_ts <- ts(GaringerOzone.monthly.components$Observed_No_Season,
  start = c(2010, 1), frequency = 12)

non_seasonal_trend <- MannKendall(non_seasonal_ts)
summary(non_seasonal_trend)

## Score = -1179 , Var(Score) = 194365.7
## denominator = 7139.5
## tau = -0.165, 2-sided pvalue =0.0075402

```

Answer: Similar to the Seasonal Mann-Kendall test for this data, we can reject the null hypothesis that the data is ozone concentration data is randomly ordered and accept the alternative hypothesis that mean ozone concentration has exhibited statistically significant change over time. Like the previous test, the tau and score values indicate a downward trend in mean ozone concentrations over time. Though the results of the test are slightly different, the conclusions that can be drawn from this data using both tests are ultimately the same.